



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification
for

Vehicle Service & Maintenance Management System

Version 1.0 approved.

Prepared by:

Priya K L - PES1UG24CS600

Prajwal S J - PES1UG24CS594

Harshith S - PES1UG24CS574

Basavaraj Mruthyunjaya Mudakammanavar - PES1UG24CS561

PES University,

01/10/2026



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Table of Contents

Table of Contents	ii
Revision History	iii
1. Introduction	1
1.1 Purpose	1
1.2 Intended Audience and Reading Suggestions	1
1.3 Product Scope	1
1.4 References	2
2. Overall Description	2
2.1 Product Perspective	2
2.2 Product Functions	3
2.3 User Classes and Characteristics	3
2.4 Operating Environment	4
2.5 Design and Implementation Constraints	4
2.6 Assumptions and Dependencies	4
3. External Interface Requirements	5
3.1 User Interfaces	5
3.2 Software Interfaces	5
3.3 Communications Interfaces	6
3.4 Hardware Interfaces	6
4. Analysis Models	6
5. System Features	12
5.1 Customer and Vehicle Registration	12
5.2 Service Request Intake, Review and Validation	13
5.3 Service Work Assignment	13
5.4 Service Execution and Recording	14
5.5 Service Status Tracking, History and Search	15
5.6 User and Access Administration	16
6. Other Nonfunctional Requirements	16
6.1 Performance Requirements	16
6.2 Safety Requirements	17
6.3 Security Requirements	17
6.4 Software Quality Attributes	17
6.5 Business Rules	18
7. Other Requirements	18
Appendix A: Glossary	19
Appendix B: Field Layouts	20
Appendix C: Requirement Traceability Matrix	23



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Revision History

Name	Date	Reason For Changes	Version
Priya K L	01-Oct-2026	Initial version of the SRS for the Vehicle Service & Maintenance Management System	1.0

1. Introduction:

1.1 Purpose:

This Software Requirements Specification (SRS) defines the requirements for release 1.0 of the **Vehicle Service & Maintenance Management System**. VSMMS is a web-based application used by a vehicle service centre to register customers and their vehicles, capture and triage service requests, assign work to technicians, record the service and maintenance activity actually performed, and keep customers informed of status and history.

This SRS covers the complete release 1.0 product: customer and vehicle registration, service-request intake and validation, work assignment, service execution and recording, status tracking and search, and user/access administration. It has been written for the Requirements Engineering & UML Use-Case Modelling.

1.2 Intended Audience and Reading Suggestions:

This document is intended for the following readers:

- **Developers and architects:** to design and implement the features in Sections 3, 5 and 6; read Sections 2, 3, 5, 6 and Appendix B first.
- **Project managers:** to plan scope and priorities; read Sections 1, 2 and the priority ratings in Section 5.
- **Testers / QA engineers:** to derive test cases from the Pass/Fail acceptance criteria; read Sections 5 and 6 and Appendix C.
- **UI/UX designers:** to design screens and flows; read Sections 2.3, 3.1, 4 and 5.
- **Service centre staff (service advisors, technicians) and administrators:** to understand what the system will do for their role; read Sections 1, 2 and 5.
- **Course evaluators:** to assess the requirements and models; read the whole document, especially Section 4 (Analysis Models) and Section 5.

→**Organisation of this SRS:** Section 1 introduces the document. Section 2 gives an overall description of the product, its users and its constraints. Section 3 lists the external interface requirements. Section 4 contains the analysis models: the UML use-case diagram, the use-case flow specification, the activity diagram and the entity-relationship diagram. Section 5 organises the functional requirements (FR-001 to FR-006) by system feature. Section 6 states the non-functional requirements (NFR-001 and NFR-002), quality attributes and business rules. Section 7 lists other requirements. The appendices provide a glossary, field layouts and a requirement traceability matrix.

1.3 Product Scope:

The Vehicle Service & Maintenance Management System digitises the front-of-house and workshop workflow of a vehicle service centre. A service advisor registers walk-in or returning customers and their vehicles; a customer or advisor raises a service request describing the work needed; the advisor reviews and validates the request; once valid, the work is assigned to a service technician, who performs and records the service, logs maintenance details and marks the work complete. The system keeps the service status and a full service history visible to the customer and searchable by staff.

Objectives and goals:

- Replace paper job cards and spreadsheets with a single, auditable digital record per vehicle and per service request.
- Reduce turnaround time by routing requests to the right technician quickly and by validating requests before work starts.
- Give customers self-service visibility into the status and history of their vehicle's servicing.
- Maintain accurate maintenance records that support warranty, resale and regulatory needs.

1.4 References:

The following documents and web resources are referred to in this SRS.

Ref	Title	Author / Publisher	Version / Date	Source / Location
[1]	Software Requirements Specification (SRS) Template	Adapted from K. Wiegers; supplied by PES University	As supplied, 2026	Course material (SRS-Template.docx)
[2]	Project diagrams: UML Use-Case Diagram, ER Diagram, Activity Diagram for the Vehicle Service & Maintenance Management System	Project team	2026	Supplied diagram images (Sections 4.1-4.4 of this SRS)
[3]	ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes: Requirements engineering	ISO / IEC / IEEE	2018	https://www.iso.org/standard/72089.html
[4]	IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications	IEEE	1998	IEEE Xplore
[5]	Software Requirements, 3rd Edition	K. Wiegers and J. Beatty, Microsoft Press	2013	Book
[6]	OMG Unified Modeling Language (UML) Specification	Object Management Group	Version 2.5.1, December 2017	https://www.omg.org/spec/UML/2.5.1
[7]	OWASP Application Security Verification Standard (ASVS)	OWASP Foundation	Version 4.0.3, 2021	https://owasp.org/www-project-application-security-verification-standard/
[8]	The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)	Government of India	2023	https://www.meity.gov.in

2. Overall Description:

2.1 Product Perspective:

The Vehicle Service & Maintenance Management System is a new, self-contained product; it does not replace an existing system and is not a member of an existing product family. It is conceived as a browser-based application used inside a single service centre (or a small chain) and does not depend on external financial or inventory systems in release 1.0.

The major logical components of the system and their responsibilities are:

Component	Responsibility
Web Client (responsive UI)	Browser-based interface used by Customers, Service Advisors, Technicians and Administrators.
Customer & Vehicle Registry	Registration and maintenance of customer and vehicle records (FR-001).
Service Request Service	Intake, review and validation of service requests (FR-002).
Work Assignment Service	Assigns validated requests to technicians and tracks assigned work (FR-003).
Service Execution & Recording Service	Records service activity, maintenance details and completion information (FR-004).
Status & History Service	Tracks status transitions, service history and search across records (FR-005).
User & Access Administration Service	User accounts, roles and system configuration (FR-006).
Relational Database	Persistent store for customers, vehicles, service requests, activities and maintenance records.

2.2 Product Functions:

At a high level the product shall allow users to perform the following major functions (details are given in Section 5):

- **Customer and vehicle registration:** register and maintain customer and vehicle records (FR-001).
- **Service request intake and validation:** submit, review and validate service requests, rejecting invalid ones with a clear error message (FR-002).
- **Work assignment:** assign a validated request to a technician, who can view assigned work (FR-003).
- **Service execution and recording:** perform the service, record service activity, maintenance details and completion information (FR-004).
- **Status tracking, history and search:** update and view service status, view service history and search records (FR-005).
- **User and access administration:** manage user accounts, roles/access and system configuration (FR-006).

2.3 User Classes and Characteristics:

User class	Description and characteristics	Frequency of use	Technical expertise	Privilege level	Importance
Customer / Vehicle Owner	Submits service requests for an owned vehicle and checks status and service history. Occasional use, around each service visit.	A few times a year	Basic web skills	Standard (own records only)	Most important (primary)
Service Advisor / Service Staff	Registers customers and vehicles, creates and manages service requests, reviews and validates them, assigns work and updates status. Front-of-house staff.	Many times a day	Moderate	Standard staff	Most important (primary)
Service Technician	Views assigned work, performs the service, records service activity, maintenance details and completion information.	Many times a day	Moderate; high mechanical/do main expertise	Standard staff	Most important (primary)
Administrator	Manages user accounts, access/roles and system configuration.	Occasional	Moderate to high	Elevated (admin)	Important (secondary)

2.4 Operating Environment:

- **Client:** any device with a current web browser: Google Chrome, Microsoft Edge, Mozilla Firefox and Apple Safari (latest two major versions); responsive layout for screen widths from 360 px (phones) upwards, since advisors and technicians may use tablets in the workshop.
- **Server:** Linux-based server or container platform (for example Ubuntu Server 22.04 LTS or later) behind an HTTPS reverse proxy.
- **Database:** a relational database that supports ACID transactions (for example PostgreSQL or MySQL 8+).
- **Network:** the workshop network and the public Internet; no special hardware is required beyond standard computers, tablets or phones.

2.5 Design and Implementation Constraints:

- **Role-based workflow:** the design must enforce that only a Service Advisor can validate/assign a request and only the assigned Technician can record completion for that request.
- **Standards:** all external communication shall use HTTPS with TLS 1.2 or higher.
- **Technology:** open-source frameworks and libraries only, because of the academic project budget; a modular design with separately testable registration, request, assignment and recording modules.
- **Language:** the user interface is English-only in release 1.0, but text must be externalised to allow later translation.
- **Schedule and team:** the project is developed by a small student team within one academic term; design conventions follow the team's coding standard and version control on GitHub.

2.6 Assumptions and Dependencies:

Assumptions (the project may be affected if any of these is wrong):

- Every registered customer and staff member has a valid e-mail address or phone number for contact.
- One service request concerns one vehicle; a vehicle may have many service requests over time.
- A service request is validated by a Service Advisor before it is assigned to a Technician.
- The service centre starts with a modest volume: a few hundred vehicles and at most 50 concurrent staff/customer users.

Dependencies:

- A hosting environment that offers TLS certificates and automated database backups.
- Third-party open-source libraries for the web framework, database access and password hashing.

3. External Interface Requirements:

3.1 User Interfaces:

The system provides a responsive web interface. The logical characteristics of the user interface are as follows; the detailed screen design is documented in a separate user-interface specification.

Screen / component	Used by	Purpose
Sign-up / Log-in	All users	Register or authenticate.
Customer & Vehicle Registration	Service Advisor	Register a customer and their vehicle(s); edit customer/vehicle details.
My Vehicles	Customer	View registered vehicles and their information.
Submit / Create Service Request	Customer, Service Advisor	Describe the work needed for a vehicle and submit the request.
Review Requests	Service Advisor	Review pending requests; validate or reject with a reason.
Assign Work	Service Advisor	Assign a validated request to an available technician.
My Assigned Work	Service Technician	List of requests assigned to the logged-in technician.
Service Execution / Recording	Service Technician	Record service activity, maintenance details and mark completion.
Service Status & History	Customer, Service Advisor	Track current status and the full history of a vehicle's services.
Search Records	Service Advisor, Administrator	Search customers, vehicles and service requests by various criteria.
User & Access Management	Administrator	Create/edit users, assign roles and configure system settings.

GUI standards and constraints:

- A common header on every screen shows the logo, the logged-in user's name/role and a **Help** link; a common footer links to Terms of Use.

- The interface shall follow WCAG 2.1 Level AA guidelines (keyboard navigation, sufficient colour contrast, text alternatives).
- Standard buttons: Save, Cancel, Back and Help are placed consistently across screens.
- **Error messages** are shown inline next to the field concerned, in plain language, without technical details (for example “Request rejected: vehicle registration number not found”).
- Screen layouts must work without horizontal scrolling on screens 360 px wide and above.

3.2 Software Interfaces:

Interface (name / version)	Direction	Data items or messages	Purpose and nature of communication
Relational database (e.g. PostgreSQL / MySQL 8+)	In / Out	Customer, vehicle, service-request, service-activity and maintenance-record records	Persistent storage using parameterised SQL over a TLS-protected connection.
Operating system and web server	n/a	HTTP requests and responses	Linux server running the application behind an HTTPS reverse proxy.
Libraries	n/a	Password hashing (bcrypt or Argon2), input validation	Open-source libraries; exact versions are recorded in the design document.

3.3 Communications Interfaces:

- **Web communication:** HTTPS (HTTP over TLS 1.2 or higher) between browser and server; plain HTTP requests are redirected to HTTPS.
- **Application programming interface:** internal REST-style API exchanging JSON encoded in UTF-8; timestamps are exchanged in ISO 8601 format.
- **Security and rate control:** authenticated sessions use secure, HTTP-only cookies; API requests are rate-limited to protect availability.

3.4 Hardware Interfaces:

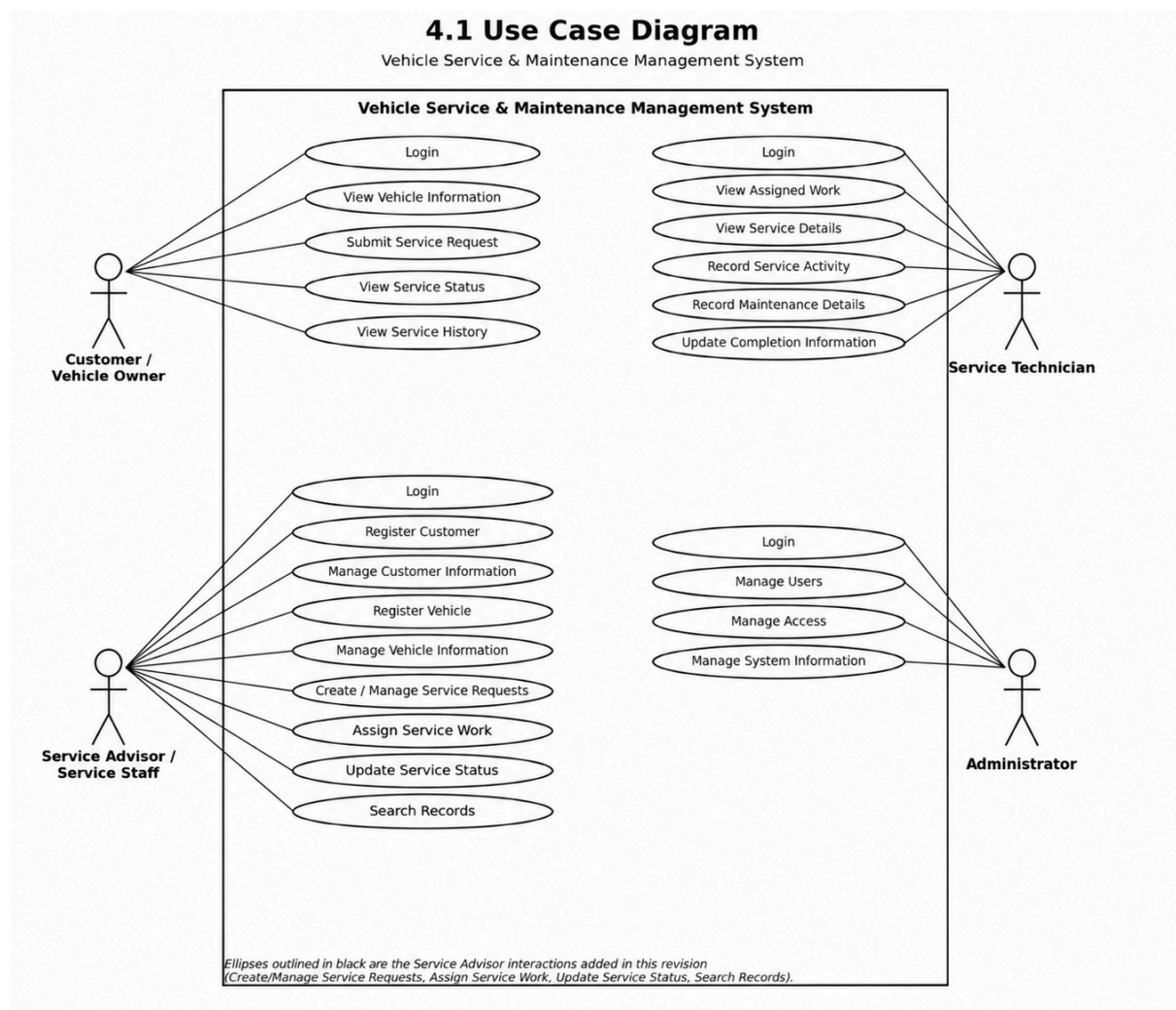
The system has no dedicated hardware interfaces. It is used only through the standard input and output devices (keyboard, pointing device or touch screen, display) of the users’ own computers, tablets or phones.

4. Analysis Models:

This section contains the analysis models of the system: the UML use-case diagram, a use-case flow specification for one core use case, the activity diagram and the entity-relationship (data) model.

4.1 UML Use-Case Diagram:

Actors: Customer / Vehicle Owner (primary), Service Advisor / Service Staff (primary), Service Technician (primary) and Administrator (secondary). The diagram below is a revision of the project's original use-case diagram: it adds four Service Advisor interactions that were present in the workflow but missing from the first draft — **Create/Manage Service Requests, Assign Service Work, Update Service Status and Search Records** — shown outlined in red. These four use cases correspond directly to FR-002, FR-003 and FR-005 below; without them the Service Advisor actor had no way to move a request from "submitted" to "assigned" to "tracked", which is the core workflow of the system.



→Use-case catalogue:

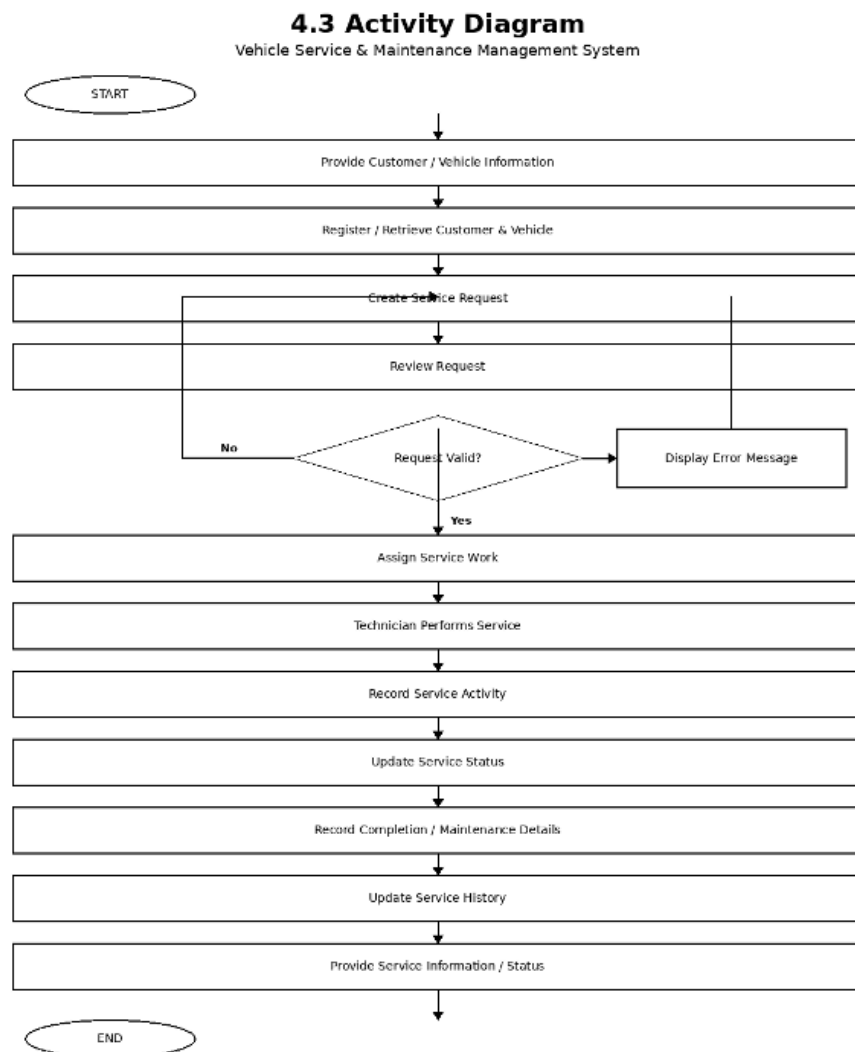
ID	Use case	Primary actor	Brief description	Related requirement
UC-01	Login	All actors	Authenticate to access role-specific functions.	FR-006
UC-02	Register Customer	Service Advisor	Create a new customer record.	FR-001
UC-03	Manage Customer Information	Service Advisor	Edit or deactivate a customer record.	FR-001
UC-04	Register Vehicle	Service Advisor	Associate a vehicle with a customer.	FR-001
UC-05	Manage Vehicle Information	Service Advisor	Edit a vehicle's details.	FR-001
UC-06	View Vehicle Information	Customer	View details of an owned vehicle.	FR-001
UC-07	Submit Service Request	Customer	Describe the work needed and submit a request.	FR-002
UC-08	Create / Manage Service Requests	Service Advisor	Create a request on a customer's behalf, or edit/withdraw one.	FR-002
UC-09	Review Request	(part of UC-08)	Review a submitted request for completeness and validity.	FR-002
UC-10	Assign Service Work	Service Advisor	Assign a validated request to a technician.	FR-003
UC-11	View Assigned Work	Service Technician	View the list of requests assigned to the technician.	FR-003
UC-12	View Service Details	Service Technician	View the full detail of an assigned request.	FR-003
UC-13	Record Service Activity	Service Technician	Log the work carried out during the service.	FR-004
UC-14	Record Maintenance Details	Service Technician	Log parts, consumables and maintenance notes.	FR-004
UC-15	Update Completion Information	Service Technician	Mark the service complete with closing notes.	FR-004
UC-16	Update Service Status	Service Advisor	Change the status of a request as it progresses.	FR-005
UC-17	View Service Status	Customer	Check the current status of a request.	FR-005
UC-18	View Service History	Customer	View the full service history of a vehicle.	FR-005
UC-19	Search Records	Service Advisor, Administrator	Search customers, vehicles and requests by criteria.	FR-005
UC-20	Manage Users	Administrator	Create, edit or deactivate user accounts.	FR-006
UC-21	Manage Access	Administrator	Assign or change roles and permissions.	FR-006
UC-22	Manage System Information	Administrator	Configure system-wide settings.	FR-006

4.2 Use-Case Flow Specification: UC-08 Create / Manage Service Requests (Review and Validation):

Use-case ID / name	UC-08 Create / Manage Service Requests (Review and Validate)
Primary actor	Service Advisor / Service Staff
Secondary actors	Customer; Service Technician (receives the assignment once validated)
Description	A service advisor reviews a submitted service request, checks that the customer and vehicle details are complete and the vehicle is registered, and either validates it (so it can be assigned, UC-10) or rejects it with a reason.
Preconditions	1. The advisor is logged in. 2. A service request exists with status Submitted/Pending. 3. The request references a registered customer and vehicle.
Postconditions (success)	1. The request status becomes Validated and is visible on the "Ready to assign" list. 2. The customer is notified that the request has been accepted.
Postconditions (failure)	The request status becomes Rejected, no work is assigned, and the customer is notified with the rejection reason.
Main success scenario	1. The advisor opens the list of Submitted service requests. 2. The advisor selects a request and opens "Review Request". 3. The system displays the request details together with the referenced customer and vehicle records. 4. The advisor checks that the request is complete (service type, description, vehicle reference) and that the vehicle is registered. 5. The system evaluates "Request Valid?": the request has all mandatory fields and a registered vehicle, so the check passes. 6. The advisor confirms validation; the system sets the request status to Validated. 7. The system makes the request available on the "Assign Service Work" screen (UC-10) and notifies the customer. The use case ends.
Alternate flow A1: request invalid (at step 5)	5a. The check fails -- for example a mandatory field is missing or the vehicle is not registered. 5b. The system displays an error message explaining why the request is invalid ("Display Error Message", matching the "No" branch of the Request Valid? decision in the activity diagram, Section 4.3). 5c. The advisor either corrects the request and resubmits it for review (returns to step 2, matching the activity diagram's loop back to "Create Service Request"), or rejects it outright with a reason, which ends the use case.
Related requirements	FR-001, FR-002, FR-003

4.3 Activity Diagram:

The activity diagram traces the end-to-end workflow from vehicle intake to closing out a service, including the validation decision described in Section 4.2.

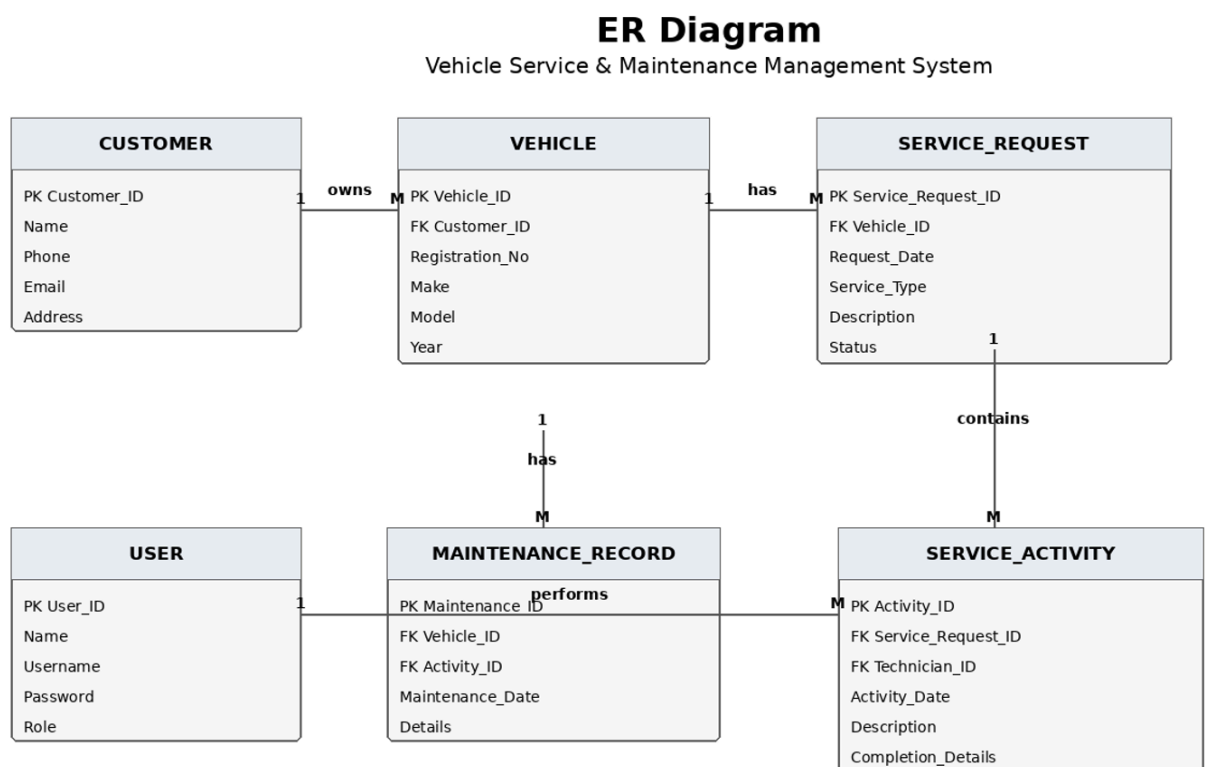


Workflow (matching the decision shown in the diagram):

1. Start.
2. Provide Customer / Vehicle Information.
3. Register / Retrieve Customer & Vehicle.
4. Create Service Request.
5. Review Request.
6. **Decision -- "Request Valid?"**
7. **If No:** Display Error Message, then loop back to Create Service Request so the request can be corrected and resubmitted.

8. **If Yes:** continue to Assign Service Work.
9. Technician Performs Service.
10. Record Service Activity.
11. Update Service Status.
12. Record Completion / Maintenance Details.
13. Update Service History.
14. Provide Service Information / Status.
15. End.

4.4 Entity-Relationship (ER) Diagram:



→**Entities and relationships:**

Entity	Key attributes	Relationship
Customer	Customer_ID (PK), Name, Phone, Email, Address	One Customer owns many Vehicles (1:M "owns").
Vehicle	Vehicle_ID (PK), Customer_ID (FK), Registration_No, Make, Model, Year	One Vehicle has many Service_Requests and many Maintenance_Records (1:M "has", both relationships).
Service_Request	Service_Request_ID (PK), Vehicle_ID (FK), Request_Date, Service_Type, Description, Status	One Service_Request contains many Service_Activity entries (1:M "contains").
Service_Activity	Activity_ID (PK), Service_Request_ID (FK), Technician_ID (FK), Activity_Date, Description, Completion_Details	One User (Technician) performs many Service_Activity entries (1:M "performs").
Maintenance_Record	Maintenance_ID (PK), Vehicle_ID (FK), Activity_ID (FK), Maintenance_Date, Details	Links a Vehicle to the Service_Activity that produced the maintenance entry.
User	User_ID (PK), Name, Username, Password, Role	Represents staff (Service Advisor, Technician, Administrator) who perform or record activities.

5. System Features:

The functional requirements are organised by system feature, that is, by the major services that the product provides. The product has six functional requirements, FR-001 to FR-006, one for each feature below. Each requirement carries a unique identifier, its type, priority, description, Pass/Fail acceptance criteria and rationale. Priorities are High, Medium or Low; the optional component ratings (Benefit, Penalty, Cost, Risk) use a relative scale from 1 (low) to 9 (high).

5.1 Customer and Vehicle Registration:

5.1.1 Description and Priority

The Service Advisor registers customers and their vehicles and keeps these records up to date; the Customer can view their own vehicle information. This is the foundation that every service request, assignment and history record builds on.

Priority: High (Benefit 8, Penalty 7, Cost 4, Risk 3)

5.1.2 Stimulus/Response Sequences

Stimulus (user action or system event)	System response
Advisor submits a new customer record (name, phone, e-mail, address).	The system validates the mandatory fields and saves the record.
Advisor registers a vehicle against an existing customer (registration number, make, model, year).	The system checks the registration number is unique and saves the vehicle, linked to the customer.
Advisor edits customer or vehicle information.	The system updates the record and keeps a change timestamp.
Customer opens "My Vehicles".	The system shows the vehicles registered to that customer.

5.1.3 Functional Requirements:

Requirement ID	FR-001
Type	Functional
Priority	High
Description	The system shall allow a Service Advisor to register and maintain customer records and to register and maintain vehicle records linked to a customer by a unique registration number, and shall allow a Customer to view the vehicles registered to them.
Acceptance Criteria	Pass: A new vehicle with a registration number that already exists is rejected with "Vehicle already registered"; a valid registration creates the vehicle linked to the correct customer, and the customer can see it under "My Vehicles". Fail: Two vehicles are saved with the same registration number, or a vehicle is created without a linked customer.
Rationale	Every later feature (requests, assignment, history) depends on a correct customer-vehicle record; duplicate or orphaned records would corrupt the whole service history.

5.2 Service Request Intake, Review and Validation:

5.2.1 Description and Priority

A Customer or Service Advisor creates a service request for a registered vehicle; the Advisor reviews it for completeness and validity before it can be assigned to a technician. An invalid request is rejected with a clear error message rather than silently dropped.

Priority: High (Benefit 9, Penalty 8, Cost 5, Risk 5)

5.2.2 Stimulus/Response Sequences

Stimulus (user action or system event)	System response
Customer or Advisor submits a service request (vehicle, service type, description).	The system saves the request with status Submitted and adds it to the advisor's review list.
Advisor opens "Review Request".	The system shows the request together with the linked customer and vehicle details.
Advisor reviews a complete request for a registered vehicle (the Request Valid? check passes).	The system sets the status to Validated and makes it available for assignment (UC-10).
Advisor reviews a request missing a mandatory field or referencing an unregistered vehicle (the Request Valid? check fails).	The system displays an error message and keeps the request open for correction and resubmission.

5.2.3 Functional Requirements

Requirement ID	FR-002
Type	Functional
Priority	High

Description	The system shall allow a Customer or Service Advisor to submit a service request for a registered vehicle, and shall allow a Service Advisor to review each submitted request against a "Request Valid?" check (mandatory fields present and the vehicle is registered). A request that fails the check shall be rejected with a displayed error message and shall remain available to correct and resubmit; a request that passes shall be marked Validated and made available for assignment.
Acceptance Criteria	Pass: A request missing the service type is kept at Submitted status and shows "Display Error Message" with the missing field named; a complete request for a registered vehicle becomes Validated and appears on the "Assign Service Work" list. Fail: An incomplete request becomes Validated, or a valid request cannot be found on the assignment list.
Rationale	Validating requests before assignment prevents technicians from starting work on incomplete or unregistered-vehicle requests, which was the gap in the original use-case diagram that this revision closes.

5.3 Service Work Assignment:

5.3.1 Description and Priority

The Service Advisor assigns a Validated service request to an available Service Technician, who can then see it on their assigned-work list.

Priority: High (Benefit 8, Penalty 7, Cost 4, Risk 4)

5.3.2 Stimulus/Response Sequences

Stimulus (user action or system event)	System response
Advisor opens "Assign Service Work" and selects a Validated request and a technician.	The system sets the request status to Assigned, links the technician and notifies them.
Technician opens "My Assigned Work".	The system lists all requests currently assigned to that technician, most recent first.
Technician opens a specific assigned request.	The system shows the full service details (UC-12).

5.3.3 Functional Requirements

Requirement ID	FR-003
Type	Functional
Priority	High
Description	The system shall allow a Service Advisor to assign a Validated service request to a Service Technician, and shall allow that Technician to view the request on an "assigned work" list and open its full details.
Acceptance Criteria	Pass: After assignment, the request appears on the named technician's assigned-work list with status Assigned and does not appear on any other technician's list. Fail: A request not yet Validated can be assigned, or an assigned request is visible to a technician it was not assigned to.
Rationale	Clear ownership of each job prevents duplicated or missed work and gives the technician exactly the information needed to start.

5.4 Service Execution and Recording:

5.4.1 Description and Priority

The assigned Service Technician performs the service and records what was done: the service activity, maintenance details (parts/consumables and notes) and finally the completion information that closes out the job.

Priority: High (Benefit 9, Penalty 7, Cost 5, Risk 4)

5.4.2 Stimulus/Response Sequences

Stimulus (user action or system event)	System response
Technician records service activity performed against an assigned request.	The system saves the activity entry with a timestamp and technician reference.
Technician records maintenance details (parts used, notes) for the vehicle.	The system saves a maintenance record linked to the vehicle and the activity.
Technician marks the request complete with completion details.	The system sets the request status to Completed and locks further edits to the activity/maintenance entries.

5.4.3 Functional Requirements

Requirement ID	FR-004
Type	Functional
Priority	High
Description	The system shall allow the assigned Service Technician to record one or more service-activity entries and maintenance-record entries against an Assigned request, and to mark the request Completed with completion information. Entries shall be attributable to the recording technician and shall not be editable after the request is marked Completed.
Acceptance Criteria	Pass: A completed request shows its service activity, maintenance details and completion notes, each timestamped and attributed to the technician; attempting to edit an entry after completion is rejected. Fail: An activity or maintenance entry is saved without a technician reference, or can still be edited after the request is Completed.
Rationale	An accurate, tamper-evident service record is the core value of the system for warranty, resale and dispute-resolution purposes.

5.5 Service Status Tracking, History and Search:

5.5.1 Description and Priority

As a request moves through Submitted, Validated, Assigned and Completed, the Service Advisor updates its status; the Customer can check current status and the full service history of a vehicle; staff can search across customers, vehicles and requests.

Priority: High (Benefit 8, Penalty 6, Cost 4, Risk 3)

5.5.2 Stimulus/Response Sequences:

Stimulus (user action or system event)	System response
Request status changes (validated, assigned, completed, etc.).	The system updates the status and timestamps the transition.
Customer opens "View Service Status" for a request.	The system shows the current status and the date of the last update.
Customer opens "View Service History" for a vehicle.	The system lists all past service requests for that vehicle with their outcomes.
Advisor or Administrator searches by customer name, registration number or request status.	The system returns matching customers, vehicles or requests.

5.5.3 Functional Requirements

Requirement ID	FR-005
Type	Functional
Priority	High
Description	The system shall track and display the status of every service request through its lifecycle (Submitted, Validated, Assigned, Completed, Rejected), shall let a Customer view the current status and the full service history of their vehicle, and shall let a Service Advisor or Administrator search customers, vehicles and service requests by name, registration number or status.
Acceptance Criteria	Pass: A status change is reflected immediately in "View Service Status"; "View Service History" lists every past request for a vehicle in date order; a search by registration number returns the correct vehicle and its requests. Fail: A status update is not reflected, a completed request is missing from the vehicle's history, or a search misses a matching record.
Rationale	Visibility of status and history is the main customer-facing value of the system, and search is what lets staff manage daily workload across many vehicles.

5.6 User and Access Administration:

5.6.1 Description and Priority

An Administrator manages user accounts, roles/access and system-wide configuration, so that only the right people can perform the right actions.

Priority: Medium (Benefit 6, Penalty 6, Cost 3, Risk 4)

5.6.2 Stimulus/Response Sequences

Stimulus (user action or system event)	System response
Administrator creates a user account and assigns a role (Customer, Service Advisor, Technician, Administrator).	The system creates the account and restricts the user's functions to that role.
Administrator deactivates a user.	The system blocks that user's login while keeping their historical records intact.
Administrator changes a system configuration setting.	The system applies and logs the change.

5.6.3 Functional Requirements

Requirement ID	FR-006
Type	Functional
Priority	Medium
Description	The system shall allow an Administrator to create, edit and deactivate user accounts, assign roles that determine which functions a user can access, and configure system-wide settings. Role changes shall take effect on the user's next action.
Acceptance Criteria	Pass: A user assigned the Technician role cannot access "Assign Service Work" (an Advisor-only function); a deactivated user cannot log in but their past records remain visible to staff. Fail: A user can perform an action outside their assigned role, or deactivating a user deletes their historical records.
Rationale	Role-based access keeps the validation -> assignment -> recording workflow trustworthy by ensuring each step can only be performed by the actor responsible for it.

6. Other Nonfunctional Requirements:

The product has two formally numbered non-functional requirements, NFR-001 (Section 6.1) and NFR-002 (Section 6.3). Sections 6.2, 6.4 and 6.5 give supporting explanations, design goals and business rules that elaborate these two requirements; they are not additional numbered requirements.

6.1 Performance Requirements:

Requirement ID	NFR-001
Type	Performance
Priority	High
Description	Under a load of 50 concurrent users, at least 95% of search, status-lookup and request-review requests shall be answered within 2 seconds, and 99% within 5 seconds.
Acceptance Criteria	Pass: A 15-minute load test with 50 virtual users shows a 95th-percentile response time of 2 seconds or less and an error rate below 1%. Fail: The 95th-percentile response time exceeds 2 seconds, or the error rate is 1% or higher.
Rationale	Advisors and technicians use search and status screens continuously through the working day; slow responses directly slow down the workshop.

6.2 Safety Requirements:

The product does not control physical equipment, so there is no risk of physical injury from its operation. The safety concerns relate to data integrity and to protecting the service record from being altered after the fact.

- Service activity, maintenance and completion entries shall not be editable once a request is marked Completed (see FR-004).

- Every status change and record edit shall be timestamped and attributed to the user who made it.
- No external safety certification is required for release 1.0.

6.3 Security Requirements:

Requirement ID	NFR-002
Type	Security
Priority	High
Description	The system shall protect user and customer data: all communication shall use TLS 1.2 or higher; passwords shall be stored only as salted hashes using an adaptive algorithm (bcrypt or Argon2); role-based access control shall ensure a user can only perform the functions permitted for their role (Customer, Service Advisor, Technician, Administrator); and every status change or record edit shall be written to an audit log.
Acceptance Criteria	Pass: A TLS scan shows that no protocol below TLS 1.2 is accepted; no password is stored in readable form; a Customer account attempting to open "Assign Service Work" receives an "access denied" (HTTP 403) response. Fail: Any plaintext password, or any user performing an action outside their assigned role.
Rationale	The service record is the product's core asset; a single integrity or access-control failure would undermine trust in the whole system.

6.4 Software Quality Attributes:

The table lists the additional quality attributes and their targets, in order of relative preference.

#	Attribute	Target (specific and verifiable)	Priority
1	Correctness / Reliability	Status and history always reflect the latest recorded activity; no completed entry can be silently changed.	Highest
2	Security	As specified in NFR-002 and Section 6.3.	Highest
3	Usability	A first-time advisor can register a customer and vehicle and create a request in under 3 minutes without help.	High
4	Performance	95% of key requests within 2 seconds at 50 concurrent users (NFR-001).	High
5	Maintainability and testability	Modular services; every requirement traceable to a test case (Appendix C).	Medium
6	Portability	Runs on any Linux host and works in the four supported browsers.	Medium
7	Recoverability	Daily backups; target recovery point of 24 hours.	Medium

6.5 Business Rules:

The following operating principles govern who may do what and under which conditions. They are not functional requirements in themselves but imply functions that enforce them.

ID	Business rule	Enforced by
BR-1	A vehicle must be registered to a customer before a service request can reference it.	FR-001, FR-002
BR-2	A service request must be Validated by a Service Advisor before it can be assigned to a technician.	FR-002, FR-003
BR-3	An invalid request is never silently dropped: it is either corrected and resubmitted, or explicitly rejected with a reason.	FR-002

ID	Business rule	Enforced by
BR-4	Only the technician a request is assigned to may record activity, maintenance details or completion for it.	FR-003, FR-004
BR-5	Once a request is Completed, its activity and maintenance entries are read-only.	FR-004
BR-6	Only Administrators can create accounts, assign roles or change system configuration.	FR-006

7. Other Requirements:

- **Database requirements:** relational schema normalised to at least third normal form; foreign keys between customers, vehicles, service requests, service activity and maintenance records; automated daily backups.
- **Internationalisation:** English user interface in release 1.0; all texts stored in resource files; UTF-8 everywhere.
- **Legal requirements:** compliance with the Digital Personal Data Protection Act, 2023 [8] for customer personal data (consent, access, correction).
- **Accessibility:** WCAG 2.1 Level AA.
- **Documentation:** a quick-start guide for advisors and technicians and an administrator guide shall be delivered with the release.

Appendix A: Glossary

The following terms, acronyms and abbreviations are used in this SRS.

Term	Definition
Customer / Vehicle Owner	A registered person who owns one or more vehicles serviced by the centre.
Service Advisor / Service Staff	Front-of-house staff who register customers and vehicles, and review, validate, assign and track service requests.
Service Technician	Workshop staff who perform the service and record activity, maintenance details and completion information.
Administrator	A privileged user who manages user accounts, roles and system configuration.
Service request	A record describing work a customer wants done on a vehicle. Statuses: Submitted, Validated, Assigned, Completed, Rejected.
Request Valid?	The validation check performed during review: the request has all mandatory fields and references a registered vehicle.
Service activity	A logged entry describing work actually carried out during a service, attributed to the technician and timestamped.
Maintenance record	A logged entry of parts, consumables and maintenance notes linked to a vehicle and a service activity.
Completion information	The closing notes a technician enters when a service request is marked Completed.
Role-based access control (RBAC)	Restricting which functions a user can perform based on their assigned role.
ACID	Atomicity, Consistency, Isolation, Durability: the properties of a reliable database transaction.
API	Application Programming Interface.
DPDP Act	The Digital Personal Data Protection Act, 2023 of India.
FR / NFR	Functional Requirement / Non-Functional Requirement.
HTTP / HTTPS	Hypertext Transfer Protocol / its secure variant over TLS.
JSON	JavaScript Object Notation, a text format for data exchange.
REST	Representational State Transfer, an architectural style for web APIs.
SRS	Software Requirements Specification.
TLS	Transport Layer Security, the protocol that encrypts web traffic.
UC	Use Case.
UI / UX	User Interface / User Experience.
UML	Unified Modeling Language.
VSMMS	Vehicle Service & Maintenance Management System (this product).
WCAG	Web Content Accessibility Guidelines.
TBD	To Be Determined.

Appendix B: Field Layouts

An Excel sheet containing field layouts and properties/attributes and report requirements. The sheets below list the fields of the main data records, matching the entities in the ER diagram (Section 4.4). Length is the maximum number of characters (or digits); data types are String, Alphanumeric, Numeric, Date and DateTime.

Sheet 1: Customer

Field	Length	Data Type	Description	Is Mandatory
Customer_ID	10	Alphanumeric	System-generated unique customer identifier	Y
Name	60	String	Customer full name	Y
Phone	15	Alphanumeric	Contact phone number	Y
Email	100	String	Contact e-mail address	N
Address	200	String	Postal address	N

Sheet 2: Vehicle

Field	Length	Data Type	Description	Is Mandatory
Vehicle_ID	10	Alphanumeric	System-generated unique vehicle identifier	Y
Customer_ID	10	Alphanumeric	Owning customer (foreign key)	Y
Registration_No	15	Alphanumeric	Vehicle registration / licence-plate number; must be unique	Y
Make	40	String	Vehicle manufacturer	Y
Model	40	String	Vehicle model	Y
Year	4	Numeric	Manufacturing year	Y

Sheet 3: Service_Request

Field	Length	Data Type	Description	Is Mandatory
Service_Request_ID	10	Alphanumeric	System-generated unique request identifier	Y
Vehicle_ID	10	Alphanumeric	Vehicle the request concerns (foreign key)	Y
Request_Date	8	Date	Date the request was submitted	Y
Service_Type	40	String	Type of service requested, e.g. Oil Change, Brake Repair	Y
Description	500	String	Description of the problem or work requested	Y
Status	15	Alphanumeric	Submitted / Validated / Assigned / Completed / Rejected	Y

Sheet 4: Service_Activity

Field	Length	Data Type	Description	Is Mandatory
Activity_ID	10	Alphanumeric	System-generated unique activity identifier	Y
Service_Request_ID	10	Alphanumeric	Request this activity belongs to (foreign key)	Y
Technician_ID	10	Alphanumeric	Technician who performed the activity (foreign key to User)	Y
Activity_Date	8	Date	Date the activity was performed	Y
Description	500	String	Description of the work carried out	Y

Field	Length	Data Type	Description	Is Mandatory
Completion_Details	500	String	Closing notes recorded when the request is marked Completed	N

Sheet 5: Maintenance_Record

Field	Length	Data Type	Description	Is Mandatory
Maintenance_ID	10	Alphanumeric	System-generated unique maintenance-record identifier	Y
Vehicle_ID	10	Alphanumeric	Vehicle the maintenance was performed on (foreign key)	Y
Activity_ID	10	Alphanumeric	Service activity that produced this record (foreign key)	Y
Maintenance_Date	8	Date	Date of the maintenance entry	Y
Details	500	String	Parts used, consumables and maintenance notes	Y

Sheet 6: User

Field	Length	Data Type	Description	Is Mandatory
User_ID	10	Alphanumeric	System-generated unique user identifier	Y
Name	60	String	User full name	Y
Username	40	Alphanumeric	Login username	Y
Password	60	Alphanumeric	Salted hash of the password; never stored in readable form	Y
Role	20	Alphanumeric	Customer / Service Advisor / Service Technician / Administrator	Y

Report Requirements: fields to be included in each report

Service Request Report	Technician Activity Report
Service_Request_ID Vehicle Registration_No Customer Name Service_Type Request_Date Status Assigned Technician Completion Date Remarks	Technician Name Service_Request_ID Vehicle Registration_No Activity_Date Description Maintenance Details Completion_Details Status

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-001	Register and maintain customer and vehicle records	Customer & Vehicle Registry	Class: Customer, Vehicle; Seq: Register Vehicle	TBD	TC-FR001-01 to 03	ST-001
2	FR-002	Submit, review and validate service requests (Request Valid? decision)	Service Request Service	Class: Service_Request; Seq: Review Request; Activity diagram Fig. 4.2	TBD	TC-FR002-01 to 04	ST-002
3	FR-003	Assign validated requests to technicians; view assigned work	Work Assignment Service	Class: Service_Request, User; Seq: Assign Service Work	TBD	TC-FR003-01 to 03	ST-003
4	FR-004	Record service activity, maintenance details and completion	Service Execution & Recording Service	Class: Service_Activity, Maintenance_Record; Seq: Record Service Activity	TBD	TC-FR004-01 to 03	ST-004
5	FR-005	Track status, view history, search records	Status & History Service	Class: Service_Request; Seq: Update Service Status, Search Records	TBD	TC-FR005-01 to 03	ST-005
6	FR-006	Manage users, roles/access and system configuration	User & Access Administration Service	Class: User; Component: Auth, RBAC	TBD	TC-FR006-01 to 02	ST-006
7	NFR-001	95% of key requests within 2 s at 50 concurrent users	Web Client; all services	Performance test plan	TBD	TC-NFR001-01	ST-007
8	NFR-002	TLS 1.2+, hashed passwords, RBAC, audit log	Security layer; Database	Component: Auth, AuditLog	TBD	TC-NFR002-01 to 03	ST-008